Mathematical Statistics | **Exam 2020-01-09 at 14:00–19:00**
Centre for Mathematical Sciences | **MASA01 A**
Lund University | **Probability theory**

Complete solutions to all problems are required. Models should be specified, approximations and conclusions should be specified and motivated.

Problems 1, 2 and 3 can give a maximum of 10 points each. Problems 4, 5 and 6 can give a maximum of 20 points each. The maximum total on the exam is 90 points. The limit for Pass (P) is at least 45 points, and the limit for High Pass (HP) is at least 67 points.

Only the paper sheets provided by the institute in the exam room shall be used, both for solutions and for sketches/notes.

Each problem solution shall start at the top of a new paper.

Write your ID number on each paper that you hand in.

Red pencil or pen is not allowed.

Only statistical tables provided by the institute are allowed. Electronic calculators are allowed. No other tables or formula sheets are allowed.

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1. Let A, B, C be events on the probability space $(\Omega, \mathcal{F}, P)$ such that A and B are independent, and A and C are disjoint. Suppose that $P(A) = 0.3$. Calculate $P(A \cap (B^c \cup C^c))$. [10p]

2. Small light bulbs are used to manufacture a Christmas tree illumination set. The company that constructs the set tries to connect the bulbs in two different ways/connections, and makes two products: “The Christmas Light” and “The Glimmer”.

   For “The Christmas Light” the bulbs ar connected in serial (after each other). In that case the set works (lights) as long as *all* bulbs function. For “The Glimmer” the bulbs are connected in parallel. In that case the set works (lights) as long as *at least one* bulb functions. (The company says that the set “works (lights)” when the set is not completely dark.)

   Suppose that one uses n light bulbs to construct a set, and that the light bulbs life lengths $T_1, T_2, \ldots, T_n$ (unit days) are independent and $Exp(1)$ distributed. Let T_{CL} be the life length for “The Christmas Light”, and T_G the life length for “The Glimmer”.

   (a) How large does n have to be to have $P(T_G > 10) > 0.5$? [5p]

   (b) How small does n have to be to have that $P(T_{CL} > 0.01) > 0.5$? [5p]

3. Let U be a discrete random variable with pmf $f_U(1) = 0.05, f_U(2) = 0.25, f_U(3) = 0.7$. Suppose that the conditional probability mass function of V given U is

$$f_{V|U}(v|u) \quad = \quad \binom{10}{v} \left(\frac{1}{u}\right)^v \left(1 - \frac{1}{u}\right)^{10-v},$$

   for $v = 0, 1, 2, \ldots, 10$.

   (a) Calculate $P(V = 5)$. [5p]

   (b) Calculate $P(U = 2|V = 5)$. [5p]

4. Suppose U is random variable with probability mass function

$$f(u) \quad = \quad \begin{cases} \frac{1}{16}, & u = 0, \\ \frac{1}{8}, & u = -1, \\ \frac{1}{16}, & u = -2, \\ \frac{3}{4} 0.2 \cdot 0.8^{u-1}, & u = 1, 2, 3, \dots. \end{cases}$$

Define the random variable $V = U1\{U > 0\} + U^{-2}1\{U \leq 0\}$.

(a) Calculate $P(U \geq 0)$ and $E(U)$. [10p]

(b) Calculate $P(V \in [1/4, 2])$ and $E(V)$. [10p]

5. The random variable X has a strictly increasing distribution function F, such that $P(X > 0) = 0.2$.

(a) Define the random variable Y by

$$Y \quad = \quad F(X) \cdot 1\{X > 0\} - 1\{X \leq 0\}.$$

Find the density of Y and specify which measure it is a density with respect to. [10p]

(b) Let $X_1, \dots, X_n$ be i.i.d., all with the same d.f. F. How large does n have to be in order to get

$$P(\sum_{i=1}^{n} 1\{X_i > 0\} = 0) \quad < \quad 0.001?$$

[10p]

6. Let $X_1, X_2, \dots$ be an infinite sequence of independent identically distributed continuous random variables, with expectation 0, variance 1/2 and $E(X_i^4) = 1$. The partial sums V_n are defined as

$$V_n \quad = \quad \sum_{i=1}^{n} X_i^2,$$

for any fixed integer n.

(a) Calculate $E(V_n)$ and $Var(V_n)$. [5p]

(b) Prove that

$$n^{-1} V_n \quad \stackrel{L^2}{\to} \quad c,$$

as $n \to \infty$, for some constant c, and find the numerical value of c. [8p]

(c) Find a numerical approximation for $P(V_{100} > 60)$. Motivate the approximation (for instance by referring to a theorem). [7p]